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Context Tuning for In-Context Optimization

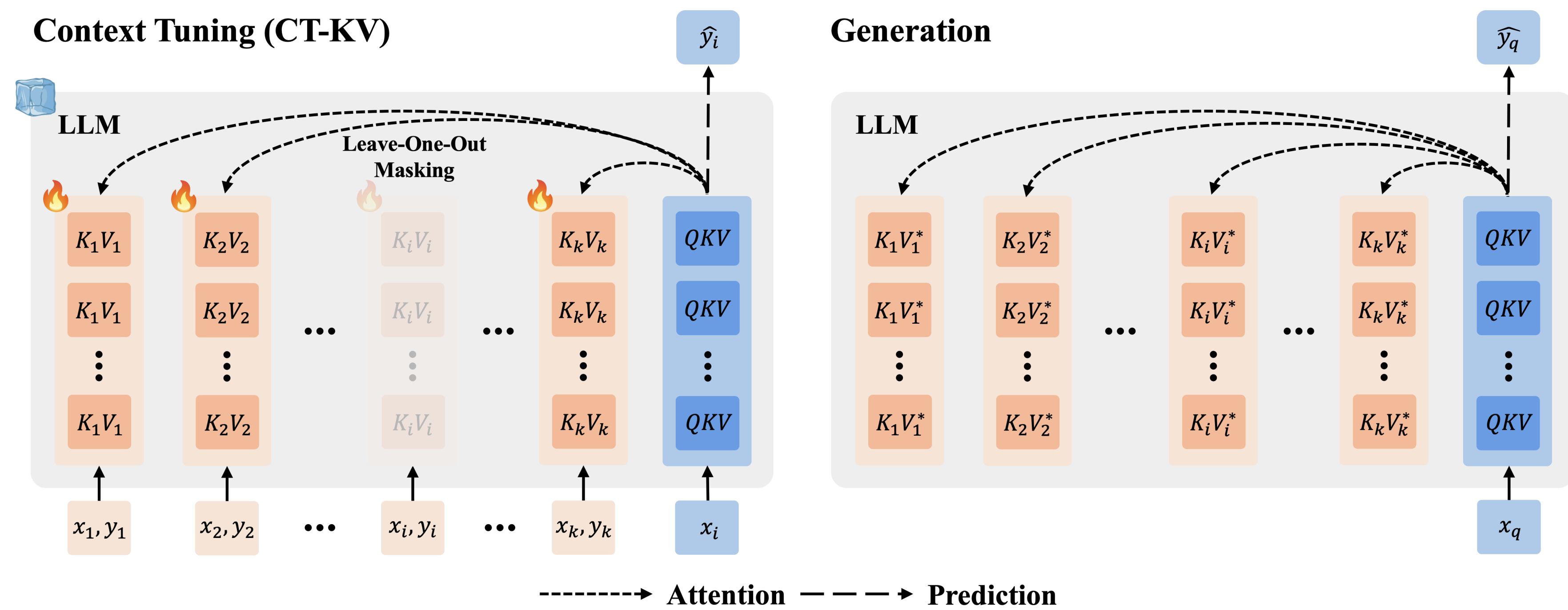
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ICML
International Conference
On Machine Learning

Optimizing the Memory Representation on the Fly

CT-KV directly optimizes the KV-prefix built from task demonstrations, before using it for generation.

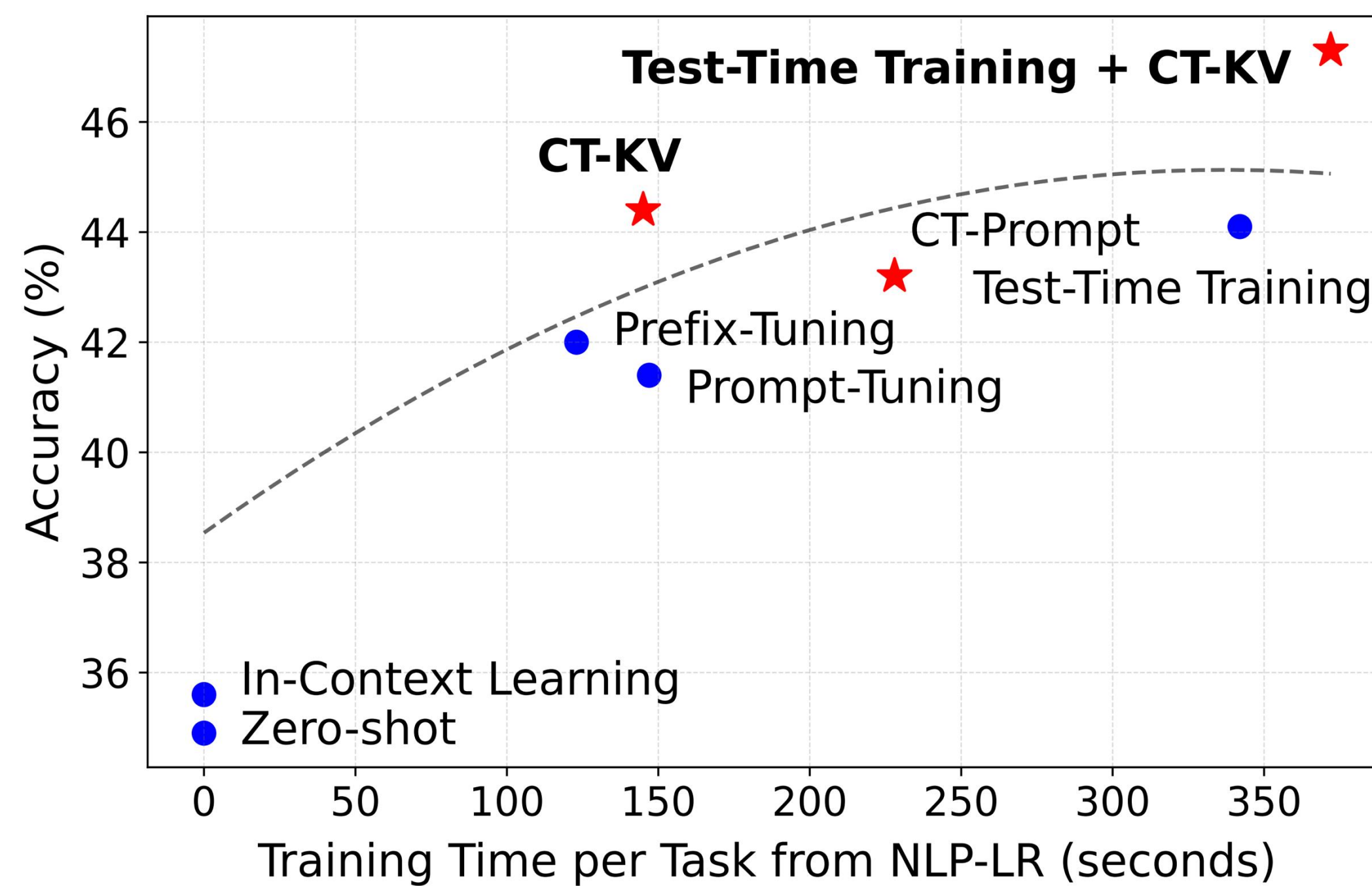


Efficient Test-Time Adaptation Across Benchmarks

By tuning the KV-prefix, Context Tuning (CT-KV) improves accuracy with lower adaptation cost and remains compatible with weight-based adaptation.

Method	NLP-LR		MMLU		BBH		ARC	
	Acc. (%)	T (s)	Acc. (%)	T (s)	Acc. (%)	T (s)	Acc. (%)	T (s)
Baselines								
Zero-Shot	34.9 ± 0.62	0	35.8 ± 0.71	0	40.9 ± 0.43	0	1.0	0
ICL	35.6 ± 0.65	0	41.2 ± 0.57	0	50.4 ± 0.78	0	13.3	0
LoRA	42.8 ± 0.88	156	40.1 ± 0.93	16	51.7 ± 0.66	9	13.5	14
Rank-Stabilized LoRA	41.7 ± 0.84	143	38.8 ± 0.87	17	46.7 ± 1.24	8	12.8	15
DoRA	42.9 ± 0.87	161	40.3 ± 0.90	16	52.6 ± 0.86	9	13.0	15
Prompt Tuning (m = 32)	41.4 ± 1.02	147	39.2 ± 1.04	15	50.8 ± 1.59	7	12.0	13
Prompt Tuning (m = # demo)	38.8 ± 1.23	231	37.3 ± 1.23	29	47.5 ± 1.84	16	14.5	49
Prefix Tuning (m = 32)	42.0 ± 0.85	123	39.9 ± 0.94	5	52.7 ± 1.12	7	9.3	14
Prefix Tuning (m = # demo)	41.1 ± 0.89	144	38.8 ± 0.81	8	52.8 ± 1.15	9	20.5	24
TTT	44.1 ± 0.65	342	43.6 ± 0.55	30	57.8 ± 1.13	14	<u>23.8</u>	56
Our Methods								
CT-Prompt	43.2 ± 0.61	228	43.6 ± 0.67	33	56.3 ± 0.98	14	22.5	52
CT-KV	<u>44.2 ± 0.55</u>	145	<u>43.7 ± 0.54</u>	9	57.9 ± 0.78	7	<u>23.8</u>	26
TTT+CT-KV	47.6 ± 0.53	372	44.1 ± 0.38	34	58.2 ± 0.73	17	25.8	63

What if we adapt LLMs by tuning their memory representation instead of weights?

In-context learning is powerful, but only if an LLM *actually understands* its context. Let's directly optimize the **memory representation** for efficient adaptation.Test-Time
Adaptation

LLM Memory

Representation
Learning

Takeaways

- In-Context Learning is not enough: forward pass builds shallow memory representation.
- Context Tuning directly tunes the memory representation, dramatically improving accuracy without updating model weights.
- Context Tuning is 2-3 faster than TTT and can be combined with it for the best results.

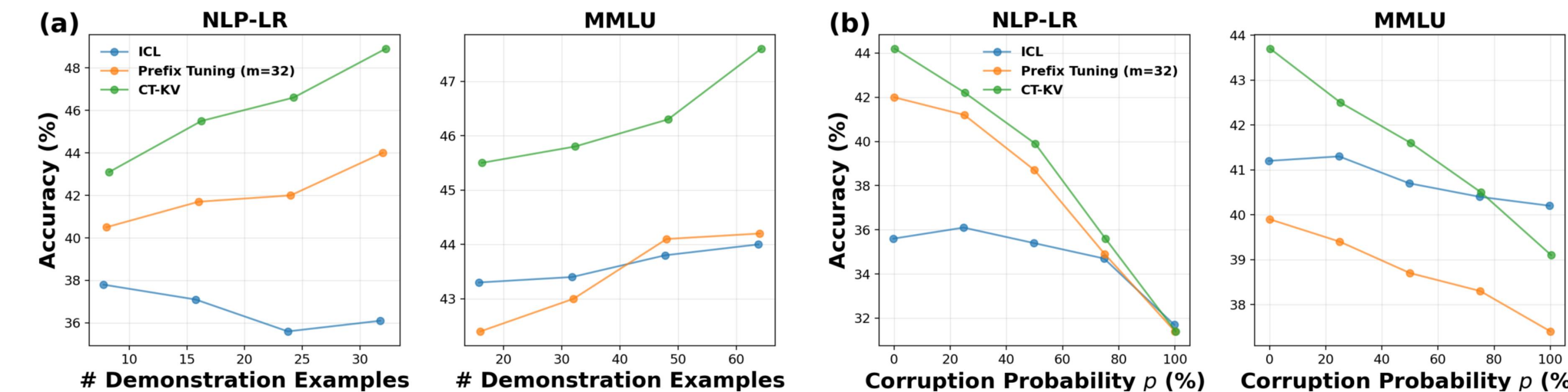
Ablation

Leave-One-Out Masking and **Token Dropout** prevent the memory representation from collapsing.

Method	NLP-LR	MMLU	BBH	ARC
Neither	41.0 ± 0.75	40.2 ± 0.73	51.4 ± 0.76	21.0
No Leave-One-Out Masking	42.6 ± 0.45	41.5 ± 0.65	54.4 ± 0.88	23.8
No Token Dropout	43.9 ± 0.62	42.7 ± 0.62	55.3 ± 0.72	21.0
Both	44.2 ± 0.55	43.7 ± 0.54	57.9 ± 0.78	22.5

Robust Gains Across Data Size, Noise, and Model Scale

CT-KV stays ahead across data sizes and remains robust to heavy label noise.



Context Tuning remains effective across modern LLMs of varying sizes and architectures.

Method	Mistral12B	DeepSeek14B	DeepSeek32B	Qwen14B	Qwen32B
ICL	51.3	55.2	63.5	60.8	63.1
Prefix Tuning (m = 32)	37.0	49.8	53.0	49.3	56.4
CT-KV	57.4	58.4	66.7	64.2	69.1

Visualizing How Memory Representation Improves

Starting from In-Context Learning output at iteration 0, CT-KV gradually corrects the ARC prediction by refining the memory representation (KV-prefix).

